NAMMA EXPERTS PU COLLEGE
2ND YEAR JEE KEY - 01.12.2023

1) 4	2) 1	3) 1	4) 3	5) 3	6) 1	7) 2	8) 2	9) 2	10) 3	11) 2	12) 2
13) 3	14) 2	15) 2	16) 3	17) 1	18) 2	19) 2	20) 2	21) 0.17	22) 1.333	23) 4.47	24) 0.5
25) 0.25	26) 0.2	27) 0.6	28) 4.0	29) 4.0	30) 0.9	31) 4	32) 2	33) 3	34) 1	35) 2	36) 3
37) 3	38) 2	39) 2	40) 1	41) 2	42) 4	43) 2	44) 1	45) 1	46) 4	47) 3	48) 1
49) 2	50) 2	51) 1.2	52) 3.0	53) 1.414	54) 8.8	55) 0.24	56) 1.155	57) 1.2	58) 16.0	59) 0.8	60) 6.0
61) 1	62) 2	63) 4	64) 1	65) 4	66) 4	67) 3	68) 4	69) 4	70) 2	71) 3	72) 4
73) 3	74) 1	75) 2	76) 4	77) 4	78) 4	79) 3	80) 1	81) 4	82) 4	83) 4	84) 4
85) 4.0	86) 2.0	87) 5.0	88) 5.0	89) 1.0	90) 5.0						

Solution :-

1)

$$n(S) = 6.6.6$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$= \frac{90}{6.6.6}$$

$$= \frac{90}{216}$$

2)

$$R.P = \left(\frac{3}{4}\right)\left(\frac{1}{2}\right)\left(\frac{3}{8}\right) + \left(\frac{1}{4}\right)\left(\frac{1}{2}\right)\left(\frac{3}{8}\right) + \left(\frac{3}{4}\right)\left(\frac{1}{2}\right)\left(\frac{1}{8}\right)$$

$$= \frac{12+9}{64}$$

$$= \frac{21}{64}$$

3)

$$P(A) = \frac{2}{3}, P(B) = \frac{3}{4}$$

$$P(A \cap B) = P(A)\,P(B)$$

$$= \frac{2}{3}\left(\frac{3}{4}\right) = \frac{1}{2}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{2}{3} + \frac{3}{4} - \frac{1}{2} = \frac{11}{12}$$

4)

$$P(A \cap B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

5)

$$\frac{{}^6c_1 . {}^4c_1}{{}^6c_1 . {}^4c_1 + {}^6c_2}$$

6)

$$P(A) = \tfrac{3}{4}, P(B) = \tfrac{4}{5}$$

$$P(E) = P(A).P\left(\overline{B}\right) + P\left(\overline{A}\right).P(B)$$

$$= \tfrac{3}{4} \times \tfrac{1}{5} + \tfrac{1}{4} \times \tfrac{4}{5} = \tfrac{7}{20}$$

7)

$$P = \tfrac{1}{2}, q = \tfrac{1}{2}$$

n = 6

$$P(x \geq 4) = P(x = 4) + P(x = 5) + P(x = 6)$$

$$= {}^6c_4\left(\tfrac{1}{2}\right)^6 + {}^6c_5\left(\tfrac{1}{2}\right)^6 = {}^6c_6\left(\tfrac{1}{2}\right)^6$$

$$= \frac{(15+6+1)}{2^6} = \frac{22}{64} = \frac{11}{32}$$

8)

$$P(A \cup B) = P(A)$$

$$B \subseteq A$$

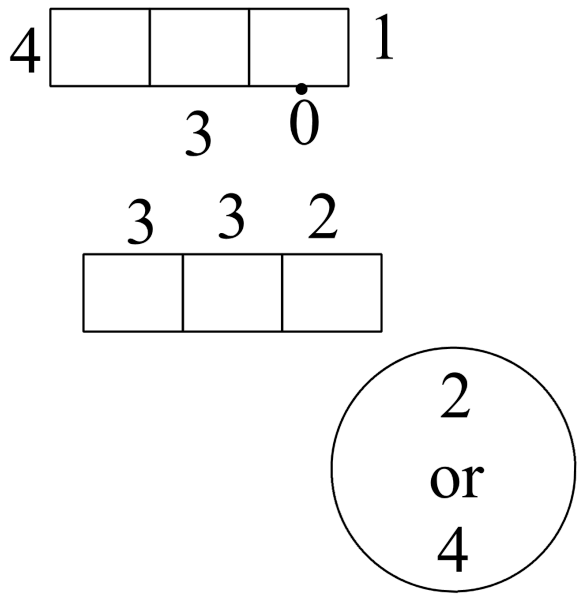
$$P(A \cap B) = P(B)$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = 1$$

9)

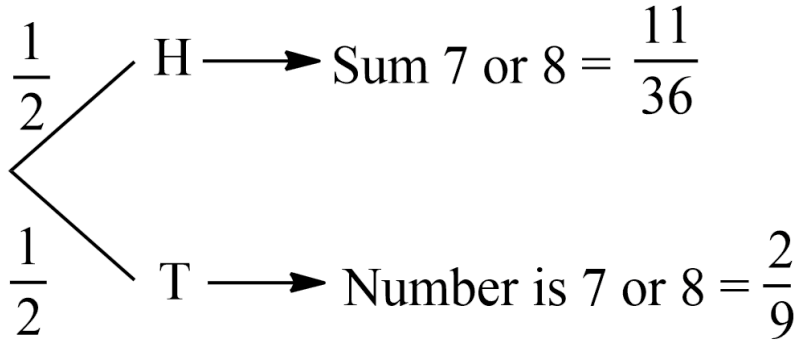
$$n(S) = 4.4.3$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{\overset{10}{\cancel{3}}\ \cancel{0}}{4.4.\ \cancel{3}} = \frac{5}{8}$$



10)

$$P(A) = \frac{1}{2} \times \frac{11}{36} + \frac{1}{2} \times \frac{2}{9} = \frac{19}{72}$$



11)

$$P(E) = \frac{{}^5c_3}{{}^9c_3}; P\left(\overline{E}\right) = 1 - \frac{{}^5c_3}{{}^9c_3}$$

$$R.P = P\left(\overline{E}\right); P(E)$$

12)

$$WWWWR$$

$$\frac{9}{10} \times \frac{8}{9} \times \frac{7}{8} \times \frac{6}{7} \times \frac{1}{6} = \frac{1}{10}$$

13)

$$V(ax \pm b) = a^2 V(X)$$

14)

$$\mu = \sum x_i P(X = x_i)$$

15)

$B_1 \to 1, 2, 3, 4$
$B_2 \to 2, 4, 6, 7, 8, 9$
$P(B_1) = \frac{1}{2}, P(B_2) = \frac{1}{2}$
E be the event getting 2 or 4
$P(E) = P(B_1).P(E/B_1) + P(B_2).P(E/B_2)$
$= \frac{1}{2} \times \frac{2}{4} + \frac{1}{2} \times \frac{2}{6}$
$= \frac{1}{4} + \frac{1}{6} = \frac{3+2}{12} = \frac{5}{12}$

16)

A={26, 62, 34, 43}

$$P(A) = 1 - \left(\frac{86}{90}\right)^3 = 1 - \left(\frac{43}{45}\right)^3$$

17)

$P\left[(E_1 \cup E_2) \cap \left(\overline{E_1} \cap \overline{E_2}\right)\right]$
$= P(E_1 \cup E_2).P\left(\overline{E_1} \cap \overline{E_2}\right)$
$= [P(E_1) + P(E_2) - P(E_1)P(E_2)]\,[(1 - P(E_1$
$= P(E_1 \cup E_2)\left[P\left(\overline{E_1 \cup E_2}\right)\right]$
Let $P(E_1 \cup E_2) = K$
$= K(1 - K)$
$f(K) = K - K^2$
$f^1(K) = 1 - 2K$
$f^1(K) = 0 \Rightarrow K = \frac{1}{2}$
Max value of $f(K) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$
$\therefore f(K) \leqslant \frac{1}{4}$

18)

n(s)=n!
R.P =1- P(all the letters go to the right place)
$= 1 - \frac{1}{n!}$

19)

A $\to$ all 5 times head(special or ordinary coin)
B $\to$ double -headed coin

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{10}}{\frac{1}{10}.1 + \frac{9}{10}\left(\frac{1}{2}\right)^5} = -$$

20)

$$Sol:\ P[A \cap (B \cup C)] = P(A)P(B \cup C)$$

$$P[A \cap (B \cap C)] = P(A)P(B \cap C)$$

21)

$$k = \frac{1}{10}, P(X > 6) = P(x = 7)$$

22)

$$P(E \cap F) = P(E).P(F) = \frac{1}{12} = xy = \frac{1}{12}$$
$$P(E \cap F) = P(E).P(F) = \frac{1}{2}$$
$$\Rightarrow (1 - P(E))(1 - P(F)) = \frac{1}{2}$$
$$\Rightarrow 1 - x - y + xy = \frac{1}{2}$$
$$1 - x - y + \frac{1}{12} = \frac{1}{2}$$
$$1 - x - y = \frac{1}{2} - \frac{1}{12} = \frac{5}{12}$$
$$x + y = \frac{7}{12}$$
$$x + \frac{1}{12x} = \frac{7}{12}$$
$$\frac{12x^2 + 1}{12x} = \frac{7}{12}$$
$$12x^2 - 7x + 1 = 0$$
$$12x^2 - 4x - 3x + 1 = 0$$
$$4x(3x - 1) - 1(3x - 1) = 0$$
$$x = \frac{1}{3}, x = \frac{1}{4}$$
$$y = \frac{1}{4}, y = \frac{1}{3}$$
$$\therefore \frac{x}{y} = \frac{1/3}{1/4} = \frac{4}{3}(or)\frac{1/4}{1/3} = \frac{3}{4}$$

23)

X=x_i	1	2	3	4	5	6
P(X=x_i)	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

Mean of $X = \mu = \sum x_i P(X = x_i)$

24)

$$A = \{(1,1)(2,2)(3,3)(4,4)(5,5)(6,6)\}$$
$$A \cap B = \{(4,4)(5,5)(6,6)\}$$
$$P(\frac{B}{A}) = \frac{P(A \cap B)}{P(A)} = \frac{n(A \cap B)}{n(A)} = \frac{3}{6} = \frac{1}{2}$$

25)

$$\sum P(X = x_i) = 1$$

26)

Required Probability =
$$\frac{1}{6}\left[\frac{{}^6c_1}{{}^{10}c_1} + \frac{{}^6c_2}{{}^{10}c_2} + \ldots\ldots + \frac{{}^6c_6}{{}^{10}c_6}\right]$$

27)

$$P(T) = \frac{2}{3}$$
$$P(F) = \frac{1}{3}$$
$$S = \{1,2,3,4,5,6,7\}$$
$$P(E) = \frac{3}{7}, P(O) = \frac{4}{7}$$
$$P(Actually\ \text{even}) = \frac{\frac{2}{3} \times \frac{3}{7}}{\frac{2}{3} \times \frac{3}{7} + \frac{1}{3}\frac{4}{7}}$$
$$= \frac{6}{10} = \frac{3}{5}$$

30)

$$P\left(\frac{B}{A \cup B}\right) = \frac{P(B)}{P(A \cup B)} = \frac{9}{10}$$

31)

[slit seperation d=0.5 mm, Distance of screen D=0.5 m]

Distance between 3rd maxima from 2nd minima on the other side

$$y = \frac{3\lambda D}{d} + \frac{(2(2) - 1)}{2}\frac{\lambda D}{d}$$
$$\Rightarrow y = \frac{9\lambda D}{2d}$$
$$y = \frac{9 \times 500 \times 10^{-9} \times 0.5}{2 \times 0.5 \times 10^{-3}}$$
$$= 2.25\ mm$$

32)

By using
$$I = 4I_0\cos^2\left(\frac{\phi}{2}\right) = 4I_0\cos^2\left(\frac{\pi\Delta}{\lambda}\right)$$
$$\left\{\because \phi = \frac{2\pi}{\lambda}\Delta\right\}$$
$$\Rightarrow \frac{I_1}{I_2} = \frac{\cos^2\left(\frac{\pi\Delta_1}{\lambda}\right)}{\cos^2\left(\frac{\pi\Delta_2}{\lambda}\right)} = \frac{\cos^2\left(\frac{\pi.\frac{\lambda}{4}}{\lambda}\right)}{\cos^2(0)} = \frac{1}{2}$$

33)

The length of the screen used portion for 15 fringes, and also for ten fringes
$$15 \times 500 \times \frac{D}{\lambda} = 10 \times \frac{\lambda D}{\lambda}$$
$$15 \times 50 = \lambda$$
$$\lambda = 750nm$$

34)

The impact of an image on the retina remains for 0.1 sec.

35)

Position of 3rd bright fringe $x_3 = \dfrac{3D\lambda}{d}$

$$\Rightarrow \lambda = \frac{x_3 d}{3D} = \frac{(0.9 \times 10^{-2}) \times (0.28 \times 10^{-3})}{3 \times 1.4} = 6000 \overset{0}{A}$$

36)

1)In case of point source,$I\,\alpha\,\frac{1}{r^2}$

2)$R.\,P\,\alpha\,\frac{1}{\lambda}$

3)$S_2P - S_1P = \text{Constant}$,represents hyperbola

4)Resultant intensity due to n identical incoherent sources is $I_R = nI_0$.

37)

For central bright $\phi = 0^0$, $I = 4I_0 \rightarrow (1)$.

If one slit is closed then no interference takes place now intensity at centre is

$I^1 = I_0 \rightarrow (2)$

$\frac{eq\ (1)}{eq\ (2)} \Rightarrow \frac{I^1}{I} = \frac{1}{4} \Rightarrow I^1 = \frac{I}{4}$

38)

$$m_\infty = -\frac{v_o}{u_o} \times \frac{D}{f_e}$$

From $\dfrac{1}{f_o} = \dfrac{1}{v_o} - \dfrac{1}{u_o}$

$$\Rightarrow \frac{1}{(+1.2)} = \frac{1}{v_o} - \frac{1}{(-1.25)}$$

$$\Rightarrow v_o = 30\ cm$$

$$\therefore |m_\infty| = \frac{30}{1.25} \times \frac{25}{3} = 200$$

39)

Convex mirror is used as a rear view mirror.

40)

Fringe visibility $= \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$

$= \frac{I_1 + I_2 = 2\sqrt{I_1 I_2} - \left(I_1 + I_2 - 2\sqrt{I_1 I_2}\right)}{I_1 + I_2 + 2\sqrt{I_1 I_2} + \left(I_1 + I_2 - 2\sqrt{I_1 I_2}\right)}$

$= \frac{4\sqrt{I_1 I_2}}{2(I_1 + I_2)}$

$= \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$

$= \frac{2\sqrt{\beta}}{1+\beta} \left(\therefore \quad \beta = \frac{I_2}{I_1}\right)$

41)

$$I = I_0 \cos^2\left(\frac{\phi}{2}\right) = K\cos^2\left(\frac{\pi}{\lambda}(\Delta x)\right)$$

where Δx is path difference

$$I = K\cos^2\left[\frac{\pi}{\lambda}\left(\frac{\lambda}{4}\right)\right] = \frac{K}{2}$$

42)

$L = f_o + f_e$

$\rightarrow 20 = f_o + f_e \rightarrow (1)$

$|m| = \dfrac{f_o}{f_e} = 9$

$\Rightarrow f_o = 9f_e$

$\rightarrow 20 = 9f_e + f_e = 10f_e$

$\rightarrow f_e = 2\,cm$

$\therefore\ 20 = f_o + 2$

$\Rightarrow f_o = 18\ cm$

43)

$M = 1 + \frac{D}{f}$

$= 1 + \frac{25}{8}$

= 1 + 3.25 = 4.25

44)

$m = -\dfrac{f_o}{f_e}$

$\rightarrow m = -\dfrac{60}{5} = -12$

45)

$\frac{1}{V} - \frac{1}{U} = \frac{1}{f}$

$\frac{1}{-(1.5)} + \frac{1}{\infty} = \frac{1}{f}$

$P = -\frac{2}{3} = -0.67\,D$

46)

$m = m_o \times m_e$

$\rightarrow 30 = m_o \times \left(1 + \dfrac{D}{f_e}\right)$

$30 = m_o\left(1 + \dfrac{25}{5}\right)$

$m_o = 5$

47)

$m = m_0 \times m_e$

$m = 25\,(6)$

$m = 150$

48)

1)For intensity maxima,

$\phi = 2\pi n\ \ then\ \ I_{\max} = 4I$

2)For intensity minima,

$\phi = (2n-1)\,\pi\ \ then\ \ I_{\min} = 0$

49)

$\dfrac{I_{\max}}{I_{\min}} = \dfrac{9}{1}$

$\dfrac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \dfrac{9}{1}$

$\dfrac{A_1 + A_2}{A_1 - A_2} = \dfrac{3}{1}$

$\dfrac{A_1}{A_2} = \dfrac{3+1}{3-1} = \dfrac{4}{2} = \dfrac{2}{1}$

50)

For intensity minima,phase difference between interfering waves $\phi = (2n-1)\pi$

Where n=1,2,3,---

51)

$$A(\mu - 1) = A^1\left(\mu^1 - 1\right)$$

$$\frac{A}{A^1} = \frac{\mu^1 - 1}{\mu - 1} = \frac{1.6-1}{1.5-1} = \frac{0.6}{0.5} = 1.2$$

52)

Given $\Rightarrow 15 = \dfrac{f_0}{f_e} \Rightarrow f_0 = 15\ f_e$

$L = f_0 + f_e$ (constant)

$L = 16\ f_e$

The focal length air

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\Rightarrow \frac{1}{f} = 0.5\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\Rightarrow \frac{1}{R_1} - \frac{1}{R_2} = \frac{R}{2f}$$

$$\Rightarrow \frac{1}{R_1} - \frac{1}{R_2} = \frac{2}{f}$$

focal length in water

$$\frac{1}{f'} = \left(\frac{\mu}{\mu_q} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\Rightarrow f' = uf_e$$

"When objective was replaced"

The new magnification

$$M' = \frac{f_0{}'}{f_e{}'},\ f_e{}' = 4f_e$$

$$L = f_0{}' + f_e{}' \Rightarrow 16f_e = f_0{}' + 4f_e$$

$$\Rightarrow f_0{}' = 12f_e$$

new magnification

$$M = \frac{f_0{}'}{f_e{}'} = \frac{12f_e}{4f_e} \Rightarrow M' = 3$$

53)

Given $i_1 = i_2 = \dfrac{3}{4}(A) = \dfrac{3}{4} \times 60^o = 45^o$

$$r_1 = r_2 = \frac{A}{2} = 30^o$$

Apply Snell's law at 1st surface

$$1 \times \sin\ 45^o = \mu \sin 30^o$$

$$\mu = \sqrt{2} = 1.414$$

54)

$$p = \frac{f_0}{f_e}\left(1 - \frac{f_e}{D}\right)$$

$$= \frac{30}{3}\left(1 - \frac{3}{25}\right)$$

$$= 10\,(1 - 0.1)$$

$$= 8.8$$

55)

Dispersion $= A\left(\mu_V - \mu_R\right)$

$= 2\,(1.62 - 1.5)$

= 0.24 radian

56)

$$i_1 = 0 \rightarrow r_1 = 0$$

$$A = 60^o$$

$$i_2 = 90^o \rightarrow r_2 = 60^o$$

$$\therefore\ \mu = \frac{\sin i_2}{\sin r_2} = \frac{1}{\frac{\sqrt{3}}{2}}$$

$$\rightarrow \mu = \frac{2}{\sqrt{3}} = \frac{2}{1.732} \approx 1.155$$

57)

Difference in number of waves = $\frac{(\mu_1 - \mu_2)t}{\lambda}$

$$400 = \frac{(\mu_1 - \mu_2)t}{5000 \times 10^{-10}}$$

$$t = \frac{400 \times 5000 \times 10^{-10}}{\left(\frac{3}{2} - \frac{4}{3}\right)}$$

$$= \frac{2 \times 10^{-4}}{\left(\frac{1}{6}\right)} = 12 \times 10^{-4}\,m$$

$$= 1.2\,mm$$

58)

$Cos\theta = \frac{n\lambda}{d} = \frac{n\lambda}{4\lambda} = \frac{n}{4}$ Where n = 0, $\pm 1, \pm 2, \pm 3, \pm 4$

$For\ n = 0\,,\ \theta = 90^0, 270^0$

$For\ n = \pm 1\,,\ Cos\theta = \pm\frac{1}{4} \Rightarrow \theta$ has four possible values.

$For\ n = \pm 2\,,\ Cos\theta = \pm\frac{1}{2} \Rightarrow \theta$ has four possible values.

$For\ n = \pm 3\,,\ Cos\theta = \pm\frac{3}{4} \Rightarrow \theta$ has four possible values.

$For\ n = \pm 4\,,\ \theta = 0^0\ \ (or)\ \ 180^0$

Total number of possible maxima=16.

59)

x=4

$$v = \frac{2\sqrt{x}}{x+1} = 0.8$$

60)

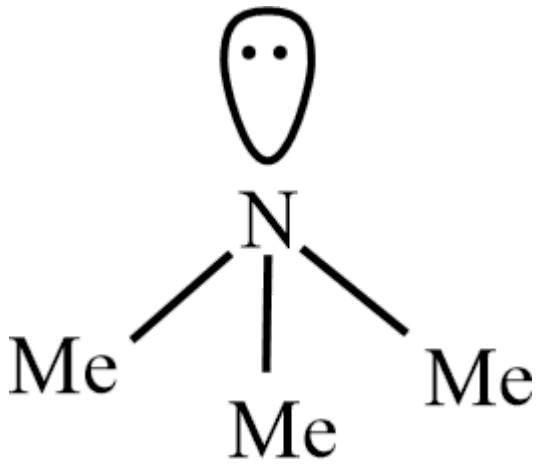
$$\phi = \pi/3,\ a_1 = 4,\ a_2 = 3$$

$$\text{So, } A = \sqrt{a_1^2 + a_2^2 + 2a_1.\,a_2 \cos\phi} \Rightarrow A \approx 6$$

61)

* Central nitrogen has 3 B.P & 1 I.P, so it is sp^3 hybridised
* Bond angle is less than tetrahedral bond angle i.e. 108°
* Central atom with 3 B.P & 1 I.P has pyramidal shape

There are three bonded pairs and one lone pair of electrons on nitrogen .So this atom requires four hybrid orbitals such as sp^3 . This would normally mean a tetrahedron of bond angle 109^o28 The changed bond angle to 108° suggests a pyramidal shape of the molecule.



62)

$$CH_3CN \xrightarrow[\text{Reduction}]{Na + C_2H_5OH} \underset{amine}{CH_3CH_2NH_2}$$

$$CH_3CN \xrightarrow[\text{Red}^n]{Na + C_2H_5OH} \underset{\text{amine}}{CH_3CH_2NH_2}$$

63)

Presence of electron withdrawing groups on benzene ring does not favour azocoupling.

64)

Gabriel phthalimide synthesis is used to prepare only 1^0 amines (except aromatic primary amine)

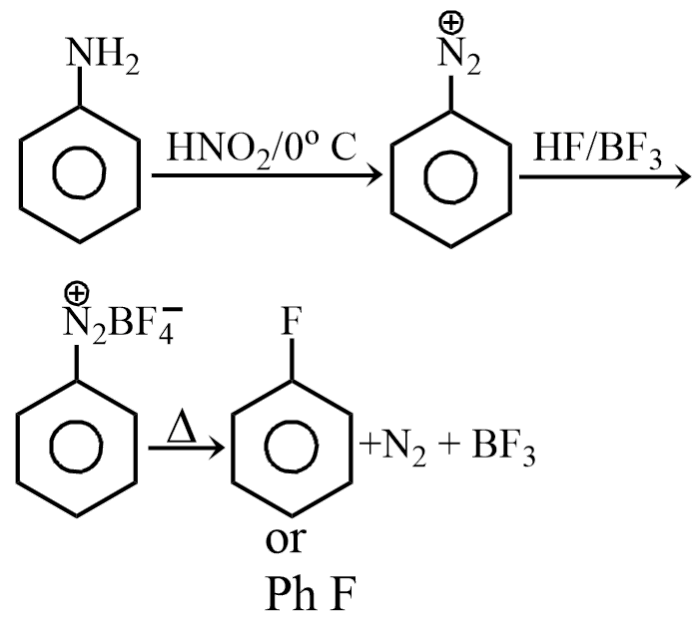
65)

Hoffmann bromamide degradation reaction

$$CH_3-\overset{\overset{\large O}{||}}{C}-NH_2 \xrightarrow{Br_2/KOH} \underset{\text{methylamine}}{CH_3NH_2}$$

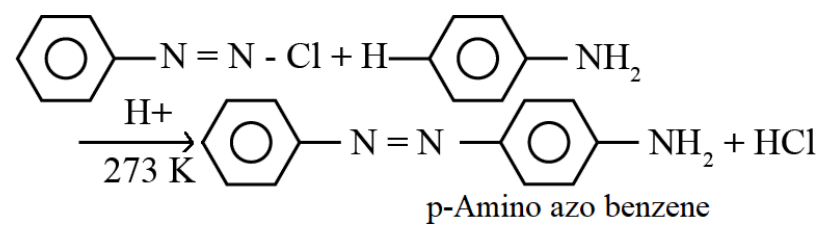
66)

Refer the figure.



67)

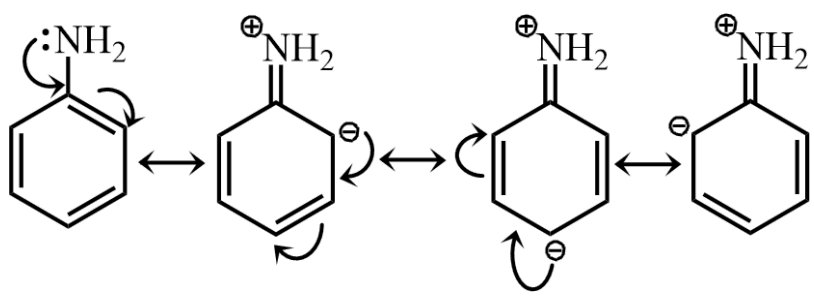
Azo dyes are derived by coupling of a phenol adsorbed on the surface of a fabric. Dyes can be prepared by diazo coupling.



68)

* In gaseous only $+I$ effect is considered to explain basic nature of amines
* Alcohols can form strong inter molecular hydrogen bond than amines because oxygen present in alcohols has higher electronegative than nitrogen present in amines
* In Aromatic amines C-N bond length is less because due to +M effect of NH_2 group partial double bond character is observed between phenyl carbon & nitrogen.

In gas phase where no solvent effects operats ,more the number of electron donating alkyl groups more basic the amine .$R_3N > R_2NH > RNH_2$ in basicity .
Oxygen is more electronegative than nitrogen . So alcohols are polar then amines. The polar alcohol is more soluble in polar water.
Due to resonance the carbon -nitrogen develops a partial double bond character. But in $RCH_2 - NH_2$ the carbon nitrogen bond has only a single bond character. Hence the bond length in aromatics is less than the bond length in alkyamines.

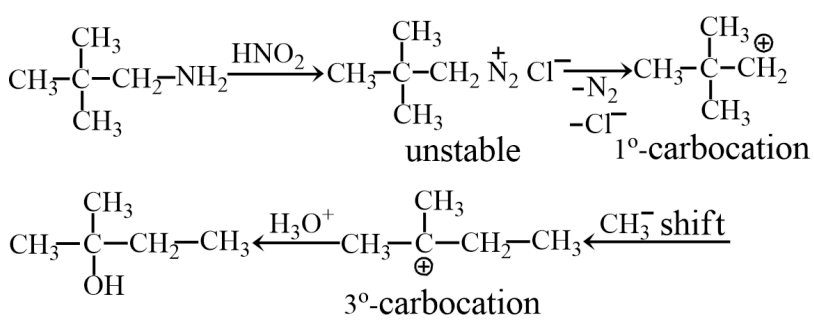


69)

Polar solvent causes poly substitution.

70)

Refer the figure.

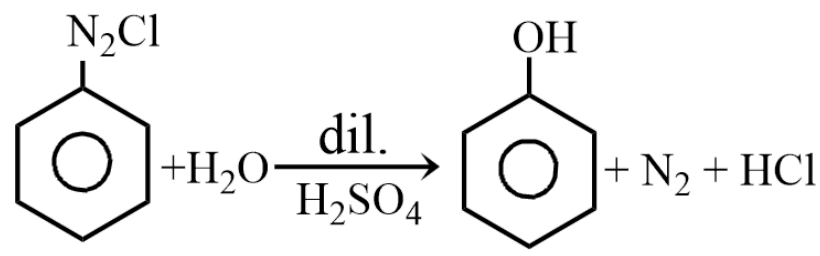


71)

tert. amines donot react with Hinsberg reagent $(C_6H_5SO_2Cl)$ due to absence of acidic hydrogen.

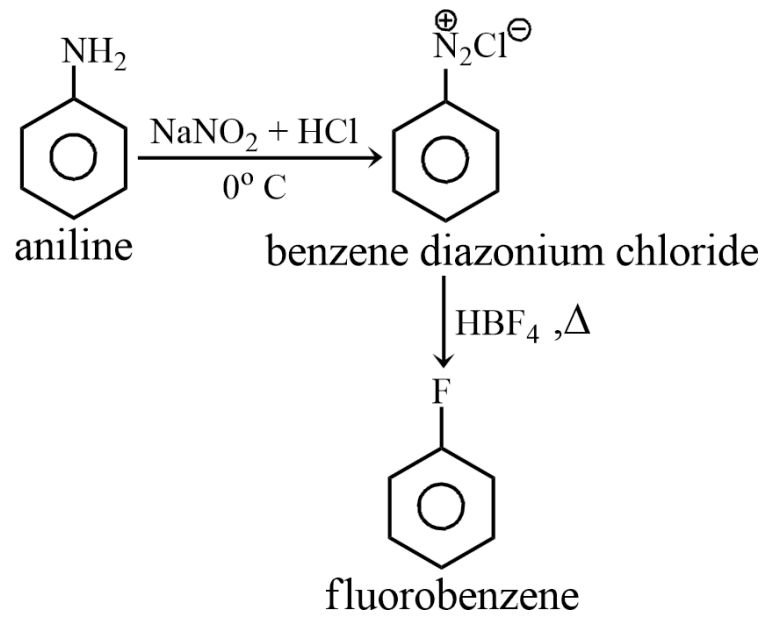
72)

Refer the figure.



73)

Refer the figure.

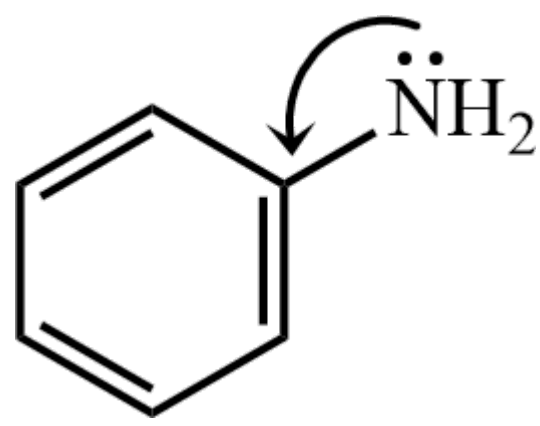


74)

Only 1°-amines are prepared by Gabriel synthesis.

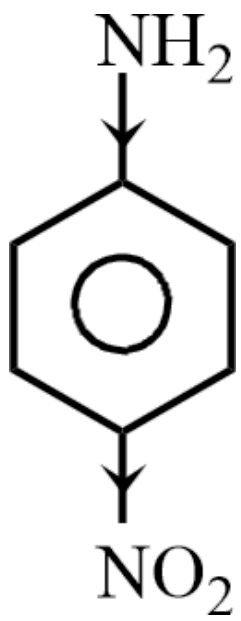
75)

In $C_6H_5NH_2$, lone pair present on nitrogen is delocalised with phenyl ring, so it is weakest base.
Alternate Solution :
An amine is basic due to the lone pairs on nitrogen. In the case of aniline, the lone pairs of nitrogen enters into resonance with the conjugated benzene part. The nitrogen electrons are delocalized and electron density on nitrogen is reduced. So aniline is the weakest base.



76)

The CH_3 group is election donating. They increase electron density on amine nitrogen and makes it more basic. But $P-NO_2$ is electron withdrawing which makes the nitrogen less basic than ammonia.



77)

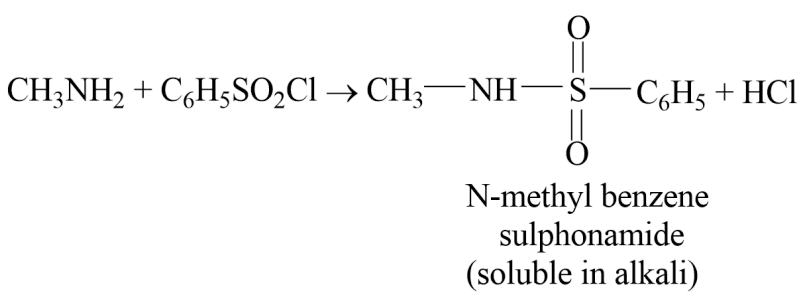
no cross product is formed in Hoffmann degradation.

78)

Unpleasant smell of carbylamine is produced when $CHCl_3 + alcoholic\ \ KOH + \rightarrow heated\ \ with$ $CH_3NH_2/C_6H_5NH_2 + CHCl_3 + 3KOH_{(alc)} \rightarrow CH_3NC + 3KCl + 3H_2O$
$[aliphatic/Aromatic\ \ i\ \ a\min e]$ or C_6H_5NC

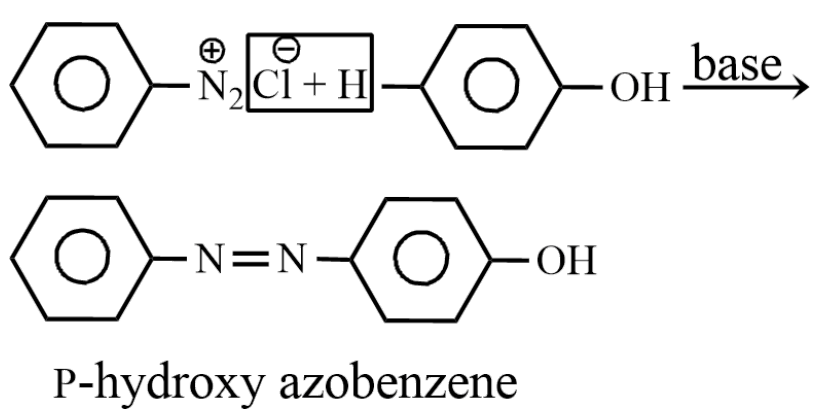
79)

Refer the figure.



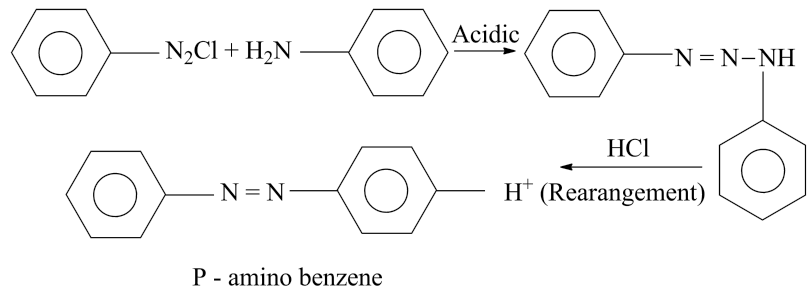
80)

Refer the figure.

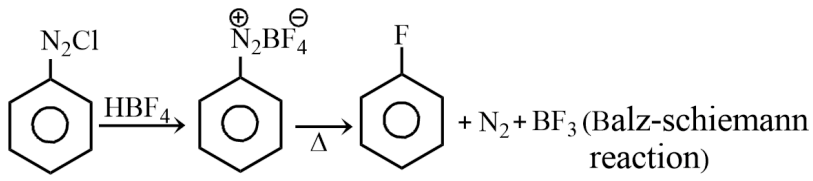


81)

if excess of HCl is used primarily to suppress the concentration of free aniline available for coupling



82)

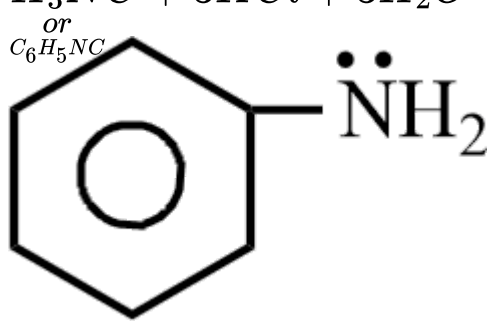


83)

$$R-NH_2 + CHCl_3 + 3KOH\,(Alc) \xrightarrow{\Delta} R-N \equiv$$

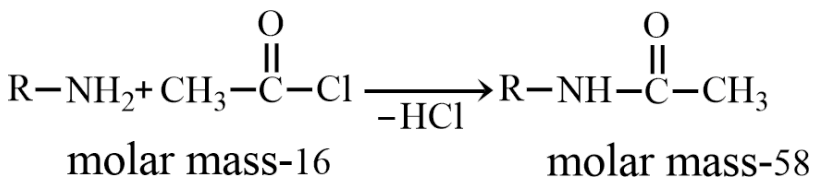
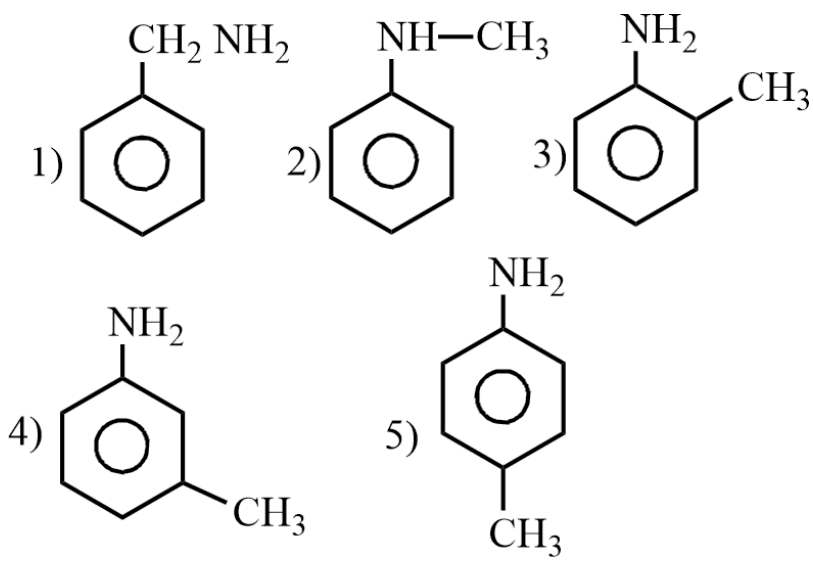
84)

The lone pair in a given structure is involved in resonance with the benzene ring and not available for donation as a Lewis base. Hence aniline is less basic than alkyl amines. Dimethylamine is a secondary amine and is most basic.



88)

Now since the molecular mass increases by 42 units as a result of the reaction of one mole of CH_3COCl with one $-NH_2$ group and the given increase in mass is 210. Hence the number of $-NH_2$ groups = 210/42=5



90)